9.Write a Java program for a simple calculator using GridLayout.

Type	Input	Expected Output	Explanation
Success	10 + 5	15	Addition
Success	20 - 8	12	Subtraction
Success	6 * 7	42	Multiplication
Failure/Edge	10 / 0	Division by zero error	Invalid operation
Failure/Edge	Operator pressed without number	Error message	UI input validation

Algorithm:

1. **Start** the program.
2. Create a JFrame with two input fields, four operation buttons (+, -, *, /), and a result field.
3. Register ActionListener for all four buttons.
4. When a button is clicked, read the two numbers from the input fields.
5. Check whether both input fields contain values.
6. Perform the selected operation:
   - + → Addition
   - - → Subtraction
   - * → Multiplication
   - / → Division
7. Check for **division by zero** before performing division.
8. Display the result in the result field.
9. If the input is invalid, display an error message.
10. **Stop** the program when the window is closed.

Program:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class SimpleCalculator extends JFrame implements ActionListener {

    JTextField num1Field, num2Field, resultField;
    JButton addBtn, subBtn, mulBtn, divBtn;

    public SimpleCalculator() {
        setTitle("Simple Calculator");
        setSize(400, 250);
```

```
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

    setLayout(new GridLayout(4, 2, 10, 10));

    add(new JLabel("First Number:"));
    num1Field = new JTextField();
    add(num1Field);

    add(new JLabel("Second Number:"));
    num2Field = new JTextField();
    add(num2Field);

    addBtn = new JButton("+");
    subBtn = new JButton("-");
    mulBtn = new JButton("*");
    divBtn = new JButton("/");

    addBtn.addActionListener(this);
    subBtn.addActionListener(this);
    mulBtn.addActionListener(this);
    divBtn.addActionListener(this);

    JPanel buttonPanel = new JPanel(new GridLayout(1, 4));
    buttonPanel.add(addBtn);
    buttonPanel.add(subBtn);
    buttonPanel.add(mulBtn);
    buttonPanel.add(divBtn);

    add(buttonPanel);

    add(new JLabel("Result:"));
    resultField = new JTextField();
    resultField.setEditable(false);
    add(resultField);

    setVisible(true);
}

@Override
public void actionPerformed(ActionEvent e) {

    try {
        if (num1Field.getText().trim().isEmpty() ||
            num2Field.getText().trim().isEmpty()) {

            resultField.setText("Error: Enter both numbers");
```

```
            return;
        }

        double num1 = Double.parseDouble(num1Field.getText());
        double num2 = Double.parseDouble(num2Field.getText());

        double result = 0;

        if (e.getSource() == addBtn) {
            result = num1 + num2;
        } else if (e.getSource() == subBtn) {
            result = num1 - num2;
        } else if (e.getSource() == mulBtn) {
            result = num1 * num2;
        } else if (e.getSource() == divBtn) {

            if (num2 == 0) {
                resultField.setText("Error: Division by zero");
                return;
            }

            result = num1 / num2;
        }

        resultField.setText(String.valueOf(result));

    } catch (NumberFormatException ex) {
        resultField.setText("Error: Invalid input");
    }
  }

  public static void main(String[] args) {
    new SimpleCalculator();
  }
}
```